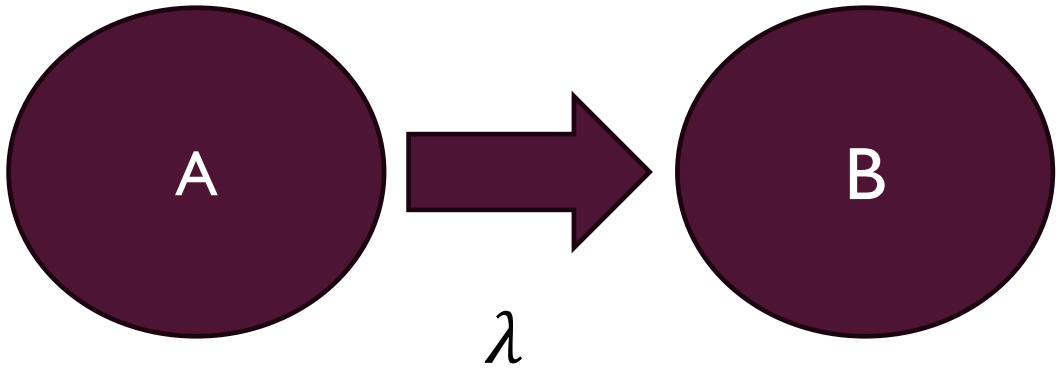
MŰFI26 – LECTURE 4.

SOME PROBLEMS THAT INVOLVE SOLUTION OF ORDINARY DIFFERENTIAL EQUATION(S)

(ONLY PROBLEMS ARE SHOWN)



RADIOACTIVE DECAY

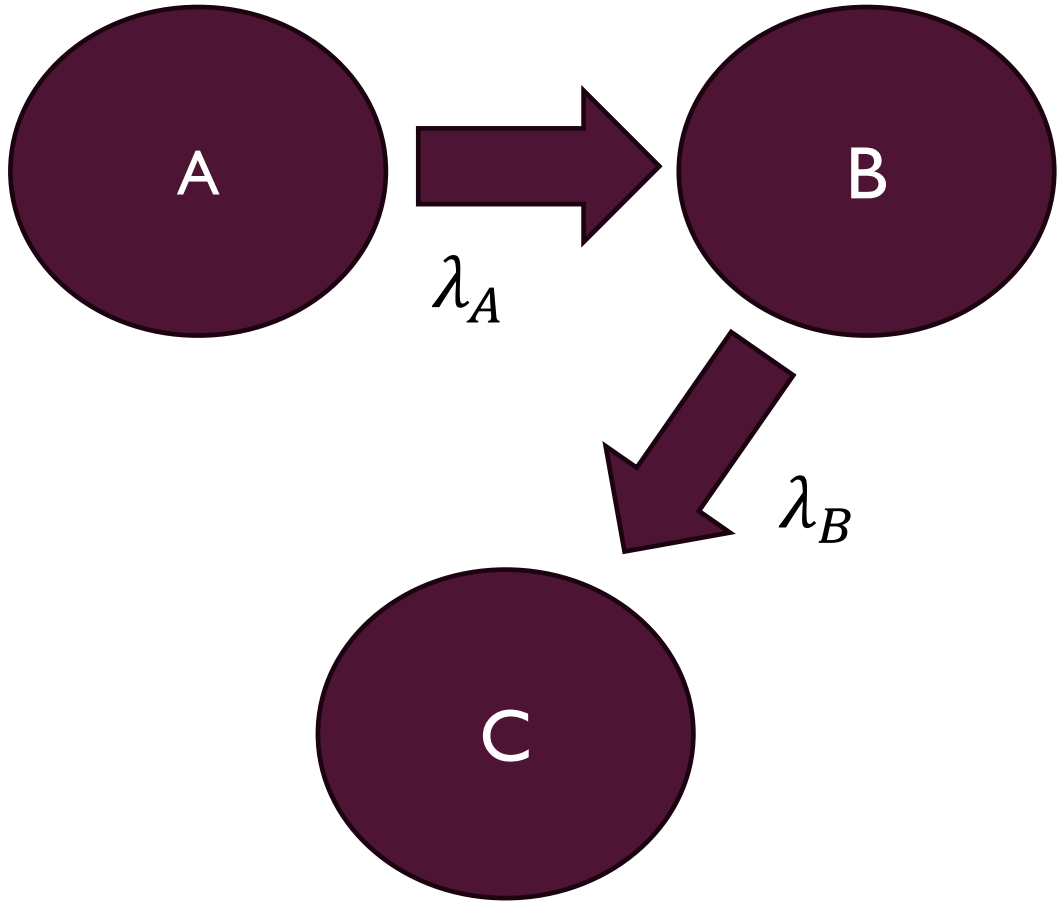


- Number of radioactive atoms decreases exponentially over time.

$$\frac{dN}{dt} = -\lambda \cdot N$$

- where N is number of atoms that not decayed so far, λ is inverse of half-life time

RADIOACTIVE DECAY – MULTIPLE DECAY



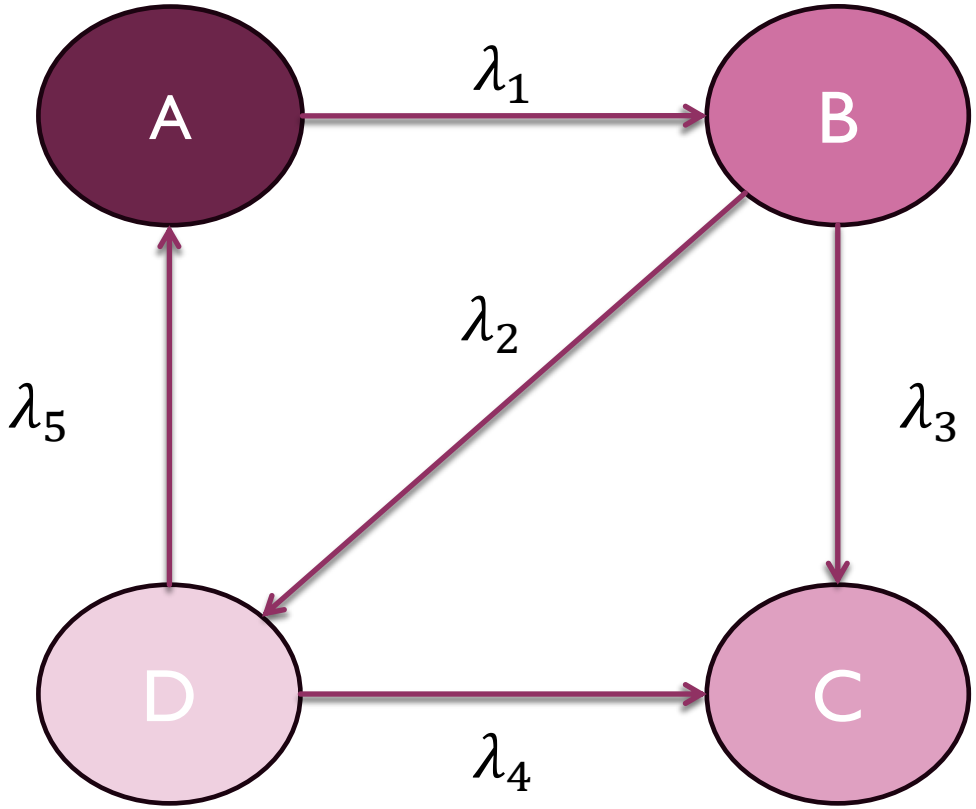
- 3 elements, 2 decay
- λ_A, λ_B are transition rates

$$\frac{dN_A}{dt} = -\lambda_A N_A$$

$$\frac{dN_B}{dt} = \lambda_A \cdot N_A - \lambda_B \cdot N_B$$

$$\frac{dN_C}{dt} = \lambda_B \cdot N_B$$

MULTIPLE DECAY – 1ST ORDER KINETIC EQUATION



- 4 elements , transition matrix
- $A = \begin{pmatrix} -\lambda_1 & 0 & 0 & \lambda_5 \\ \lambda_1 & -\lambda_2 - \lambda_3 & 0 & 0 \\ 0 & \lambda_2 & 0 & \lambda_4 \\ 0 & \lambda_3 & 0 & -\lambda_4 - \lambda_5 \end{pmatrix}$

SOLVING STRATEGY OF ODES

- using ode45, ode89 and for stiff probles ode15s
- ode45 : explicit Runge-Kutta (4,5) formula, the first one to be used
- function of DE (differential equation) : `dydt = function odefun(t,y)`
- using default values, but set relative tolerance to 10^{-6} :
  `opts = odeset('RelTol',1e-6)`
- set timespan (start time to end time) : `tspan = [0 7000];`
- set initial state : `x0 = [1000];`
- solve ODE : `[T,Y] = ode45(@odefun, Tspan, x0, opts);`
  T : time of solutions (Nx1), Y : solution of variable(s) (NxM)

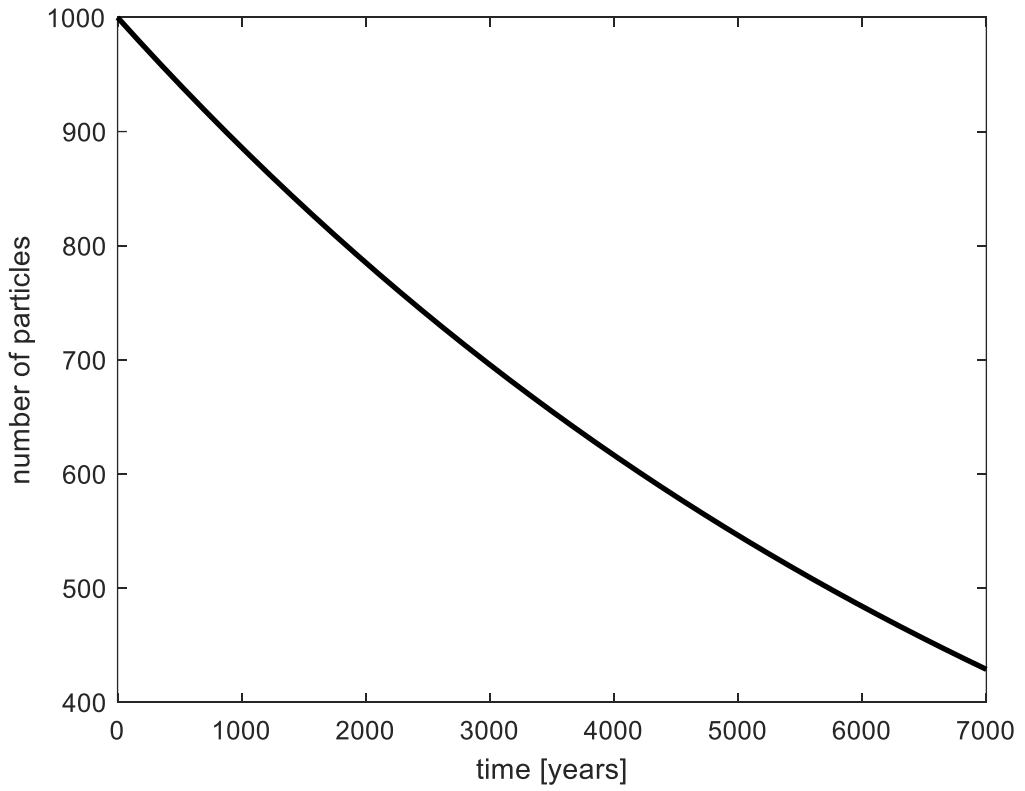
EXAMPLE – SOLUTION OF DECAY EQUATION

- Decay equation : $\frac{dN}{dt} = -\lambda \cdot N$
- set intial state : `N0 = 1000;`
- setting parameters of ODE :

```
tau_half = 5730; % in years
la = log(2)/tau_half;
```

- Solving equation using Runge-Kutta 4.5 order

```
[T,Y] = ode45(@(t,x) decayode(x,la),[0 7000],[N0]);
```



NEWTON'S LAW OF COOLING

- A hot cup of coffee cooling in the room. Temperature change is proportional to temperature difference between coffee and room's air.
- Equation that describes cooling

$$\frac{dT}{dt} = -k \cdot (T - T_{air})$$

where T is temperature of coffee, T_air is air temperature and k is heat convection constant.



CIRCUIT WITH CAPACITOR AND RESISTOR IN SERIES

- Let's use a simple circuit of capacitor, resistor and battery connected serialy. We can write equation for voltage of capacitor if battery is turned on (from 0 V to V_0 when $t = 0$)

$$\frac{dV}{dt} + \frac{V}{R \cdot C} = \frac{V_0}{R \cdot C}$$

- where V is voltage of capacitor, R and C are resistance and capacitance, V_0 is battery's voltage
- Analytical solution :

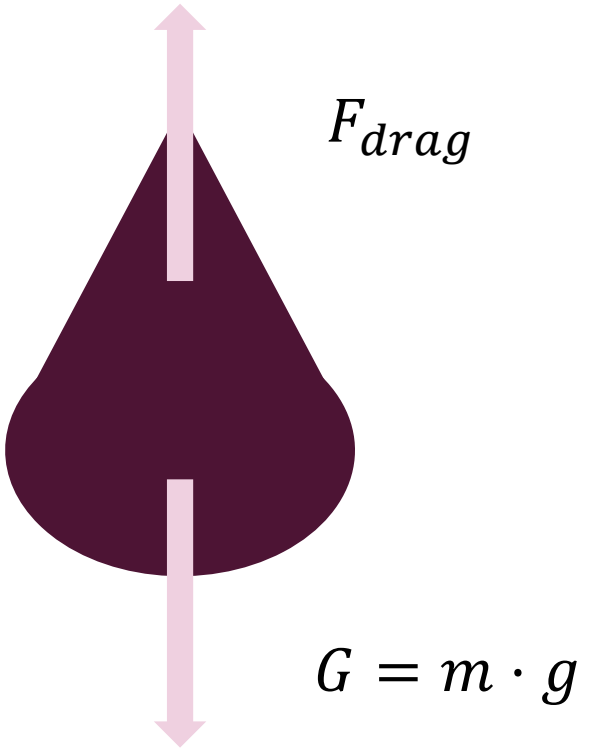
$$V(t) = V_0 \cdot \left(1 - \exp\left(-\frac{t}{R \cdot C}\right)\right), \qquad t > 0$$

DRAG FORCE OF A FALLING OBJECT

- Let us use falling problem. An object falling through air with velocity-dependent drag.
- If obkect's mass is m, and gravitational constant is m and drag force coefficient is b :

$$m \cdot \left(\frac{dv}{dt}\right) + b \cdot v^2 = m \cdot g$$

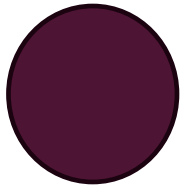
- We can calculate solution of this problem :



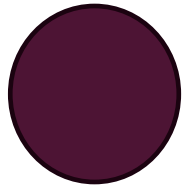
BERNOULLI EQUATIONS - POPULATION WITH LIMITED RESOURCES

- Population with limited resources (like rabbits and grass)
- Population of rabbits is denoted by P, grass (what rabbits eat) is limited and rabbits can not eat as many as they want.

$$\frac{dP}{dt} = r \cdot P \ - a \cdot P\text{^}2$$



rabbit (P)



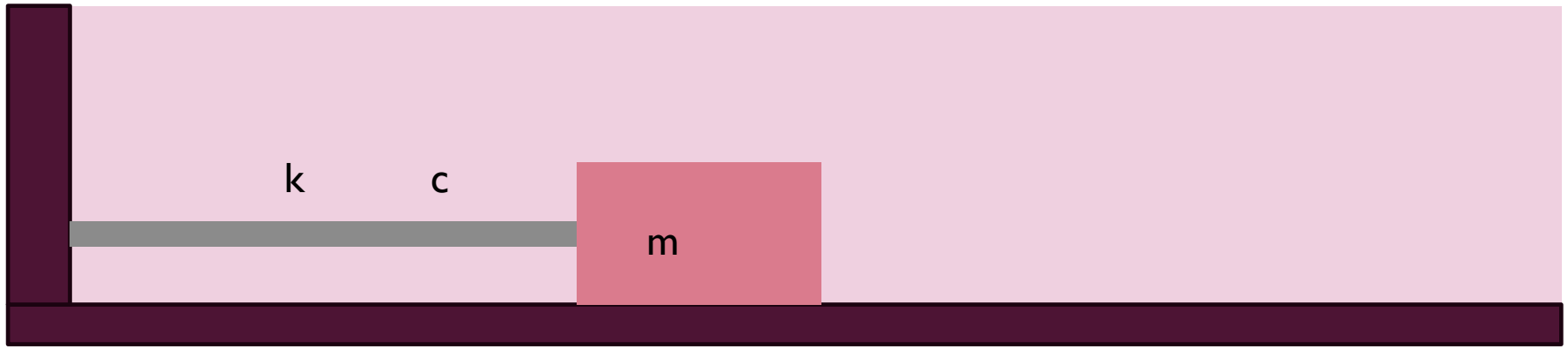
grass

OSCILLATORS' OF DIFFERENT KINDS / MECHANICAL SYSTEM

- Simplest form of a harmonic oscillator (spring-mass system) :

$$m \cdot \frac{dx^2}{dt^2} + c \cdot \frac{dx}{dt} + k \cdot x = 0$$

- is ODE of problem, where x is placement, k is spring constant, c is damping constant.
- Oscillations are not forced.

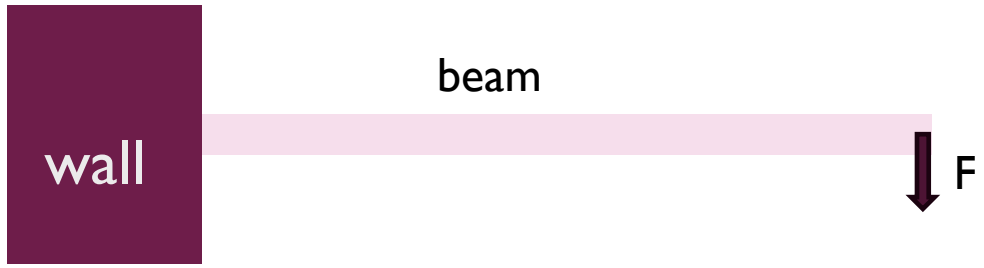


HIGHER ORDER LINEAR ODES

- Higher order ODEs are like harmonic motion (2nd derivative of space is velocity) that can be converted to set of linear equations of lower degree.
- Beam deflection : A beam is bending under load

$$E \cdot I \cdot \left(\frac{d^4 y}{dx^4}\right) = w(x)$$

- where E is Young's modulus, I moment of inertia, w is function of distributed load
- Don't forget that we need 4 boundary conditions to start.

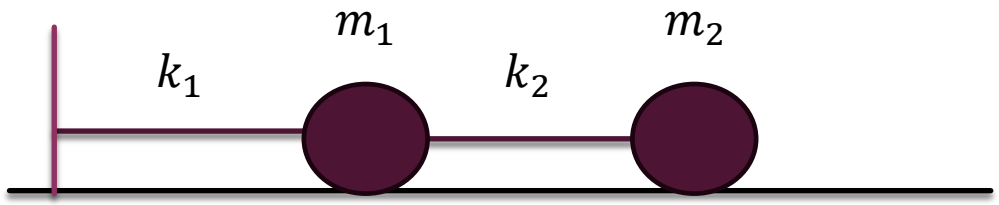


COUPLED OSCILLATIONS

- Let's connect two oscillatory systems.

$$m_1\left(\frac{d^2x_1}{dt^2}\right) = -k_1 \cdot x_1 + k_2(x_2 - x_1)))$$

- Twos masses(m_2 and m_1) are connected together, oscillations will travel across masses



$$m_2\left(\frac{d^2x_2}{dt^2}\right) = -k_2 \cdot (x_2 - x_1)$$